Department of Computer Science and Engineering

UNIT I: OPERATING SYSTEM ARCHITECTURE

1. Introduction to Operating Systems

Operating System is defined as system software managing hardware resources.

The primary goal is executing user programs and managing CPU scheduling.

1.1 Process Management

Process is defined as an instance of a computer program in execution.

Process Control Block refers to the data structure storing process context.

2. CPU Scheduling Algorithms

Turnaround Time = Completion Time - Arrival Time

Waiting Time = Turnaround Time - Burst Time

Step 1: Inspect ready queue processes.

Step 2: Allocate CPU to process with minimum burst.

Step 3: Preempt current process if shorter burst arrives.

Deadlock is defined as a situation where processes block waiting for resources.